COMPETITIVE PROGRAMMING

CP
ROADMAP

800 → 1600

From Newbie to Expert — A Complete Topic-by-Topic Guide

By Kaesarz

"Every expert was once a beginner who refused to give up."

9	60+	800	1600
RATING LEVELS	TOTAL TOPICS	FROM RATING	TO RATING

INTRODUCTION

About This Book

This roadmap is built for one purpose — to take you from Codeforces rating 800 all the way to 1600 with a clear, structured, topic-by-topic plan. No fluff. No confusion. Just exactly what you need to know, at exactly the right time.

How to Use This Book

Each chapter represents a rating milestone. The topics listed under each rating are the NEW concepts you need to learn to reach that level. All previous topics are assumed to be mastered before moving forward.

The Golden Rules of CP

01	**Practice over Theory** Reading without solving is useless. For every topic you learn, solve at least 10 problems.
02	**Upsolve Every Contest** After every contest, solve the problems you couldn't. This is where real growth happens.
03	**Read Editorials Actively** Don't just read — simulate, think, and understand WHY the solution works.
04	**Never Skip the Basics** A weak foundation will collapse under hard problems. Master each level before moving on.
05	**Consistency Wins** One hour every day beats ten hours once a week. Build the habit.

Rating → Rank Reference

Rating	Rank	Color
< 1200	Newbie / Pupil	Gray → Green
1200 – 1399	Pupil	Green
1400 – 1599	Specialist	Cyan
1600 – 1899	Expert	Blue
1900 – 2099	Candidate Master	Purple

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Contents

Rating 800

★ 800

The Foundation — Where Every CP Journey Begins

> *Master the basics. Every expert was once a beginner.*

C++ ESSENTIALS

- **Variables, Data Types, Overflow (int vs long long)** NEW
- **Conditionals & Loops** NEW
- **Functions** NEW
- **Fast I/O (ios_base::sync_with_stdio, cin.tie)** NEW

DATA STRUCTURES

- **Arrays (1D and 2D)** NEW
- **Strings (basic traversal, indexing)** NEW
- **Frequency Arrays** NEW

ALGORITHMS & TECHNIQUES

- **Sorting (std::sort, custom sort)** NEW
- **Basic Math (modulo, absolute value)** NEW
- **Brute Force** NEW
- **Basic Greedy** NEW
- **Basic Constructive** NEW

Rating 900

★ 900

STL Mastery — Your Most Powerful Weapon

> *The right data structure solves half the problem.*

STL CONTAINERS

- **Vector** NEW
- **Pair & Tuple** NEW
- **Map / Unordered Map** NEW
- **Set / Unordered Set** NEW
- **Stack** NEW
- **Queue** NEW

ALGORITHMS & TECHNIQUES

- **Prefix Sum (1D)** NEW
- **Two Pointer** NEW
- **Sliding Window** NEW
- **Intermediate Greedy** NEW
- **Intermediate Constructive** NEW

CHAPTER 3 OF 9

Rating 1000

★ 1000

Recursion & Search — Think Deeper

> “ *To understand recursion, you must first understand recursion.*

ARRAY TECHNIQUES

- **Prefix Sum (2D)** NEW
- **Difference Array** NEW

SEARCHING

- **Binary Search (basic)** NEW
- **lower_bound / upper_bound** NEW

STL ADVANCED

- **Priority Queue** NEW

RECURSION & BACKTRACKING

- **Recursion** NEW
- **Basic Backtracking** NEW

Rating 1100

★ 1100

Math & Bits — The Hidden Power

> *Mathematics is the language in which the universe is written.*

SEARCHING

- Binary Search on Answer — NEW

NUMBER THEORY

- GCD / LCM — NEW
- Prime Checking (trial division) — NEW
- Modular Arithmetic — NEW

BIT MANIPULATION

- XOR Properties — NEW
- Bit Manipulation (basic) — NEW
- __builtin_popcount — NEW

STRING ALGORITHMS

- Palindrome Check — NEW
- Anagram Check — NEW
- Intermediate Constructive (math-based) — NEW

Rating 1200

★ 1200

Number Theory & DP — Pattern Recognition

> *Dynamic programming is recursion with memory.*

ADVANCED NUMBER THEORY

- **Sieve of Eratosthenes** NEW
- **Prime Factorization** NEW
- **Divisors & Multiples Counting** NEW
- **Binary Exponentiation** NEW

DYNAMIC PROGRAMMING

- **1D DP** NEW
- **DP on Arrays** NEW
- **Coin Change Pattern** NEW
- **0-1 Knapsack** NEW
- **Subset Sum DP** NEW

CHAPTER 6 OF 9

Rating 1300

★ 1300

Graph Theory — Connecting the Dots

> *Every path starts with a single node.*

GRAPH BASICS

- **Graph Representation (Adjacency List)** NEW
- **DFS (Depth First Search)** NEW
- **BFS (Breadth First Search)** NEW
- **Connected Components** NEW
- **Cycle Detection** NEW
- **Bipartite Check** NEW
- **Flood Fill** NEW

TREES

- **Tree Basics (traversal, depth, subtree size)** NEW

GRID ALGORITHMS

- **Grid-based BFS / DFS** NEW

DYNAMIC PROGRAMMING

- **Grid DP** NEW

Rating 1400

★ 1400

Advanced Graph & DP — Level Up

> *The difference between good and great is attention to detail.*

ADVANCED GRAPH

- **Topological Sort** NEW
- **DSU (Disjoint Set Union)** NEW
- **Dijkstra Algorithm** NEW
- **Euler's Phi Function** NEW

ADVANCED DYNAMIC PROGRAMMING

- **LCS (Longest Common Subsequence)** NEW
- **LIS (Longest Increasing Subsequence)** NEW
- **Edit Distance DP** NEW

ADVANCED GREEDY

- **Interval Greedy** NEW
- **Scheduling Greedy** NEW

CHAPTER 8 OF 9

Rating 1500

★ 1500

Advanced Techniques — Specialist Territory

> *Complexity is the enemy. Master it.*

GRAPH ALGORITHMS

- **Floyd Warshall** NEW
- **Minimum Spanning Tree (Kruskal)** NEW

ADVANCED DYNAMIC PROGRAMMING

- **Digit DP** NEW
- **Bitmask Basics** NEW
- **Bitmask DP** NEW
- **DP on Trees** NEW

STACK TECHNIQUES

- **Monotonic Stack** NEW
- **Next Greater Element** NEW

SEARCH

- **Advanced Backtracking** NEW

Rating 1600

★ 1600

Expert Level — The Final Frontier

> *You are no longer a learner. You are a problem solver.*

ADVANCED DATA STRUCTURES

- **Segment Tree (basic)** NEW
- **BIT / Fenwick Tree** NEW
- **Sparse Table** NEW

ADVANCED TECHNIQUES

- **String Hashing** NEW
- **Advanced Binary Search** NEW
- **Advanced DP Optimization** NEW

FINAL CHAPTER

The Mindset of a Champion

Compete, Don't Compare
Your journey is yours alone. Never measure your progress against others. Measure it against who you were yesterday.

Embrace the Struggle
The problems that take you 3 hours to solve are the ones that teach you the most. Pain in practice means growth. Never run from hard problems.

Upsolve Without Excuses
Every problem you skip after a contest is a missed opportunity. Upsolving is not optional — it is the core of improvement.

Take Care of Your Brain
Sleep, eat well, take breaks. A tired brain solves nothing. CP is a marathon, not a sprint. Sustainable practice beats burnout every time.

Trust the Process
Rating will go up and down. That is normal. What matters is that your skill is constantly growing. Trust the process and keep showing up.

Stay Honest
Never cheat. Never copy solutions before genuinely trying. Shortcuts in CP will cost you your most valuable asset — your own ability.

"The journey of a thousand problems begins with a single line of code."

— Kaesarz